MEDBOT ZIMBABWE

Technical Architecture & Implementation Overview

ONE HEALTHCARE KNOWLEDGE SYSTEM • TWO ACCESS CHANNELS • SECURE PATIENT RECORDS

Purpose

This document presents a proposed hybrid offline-first architecture for Medbot Zimbabwe. It extends the original WhatsApp-based healthcare information and triage concept with an Android application, secure personal health records, community update points and controlled access for authorised healthcare professionals.

Status

This is a proposed design for discussion and development. The Android application, offline AI, patient-record services, synchronisation workflows and community update network should not be treated as already implemented or clinically approved.

Prepared as a technical companion to the Medbot Zimbabwe AI for Good Sandbox Hackathon submission.

1. Overall system architecture

The architecture separates user access, online AI services, verified health content, offline device capabilities, patient-record services, community distribution and human clinical oversight. WhatsApp remains the connected access channel; the Android application provides access to previously downloaded approved information when connectivity is unavailable.

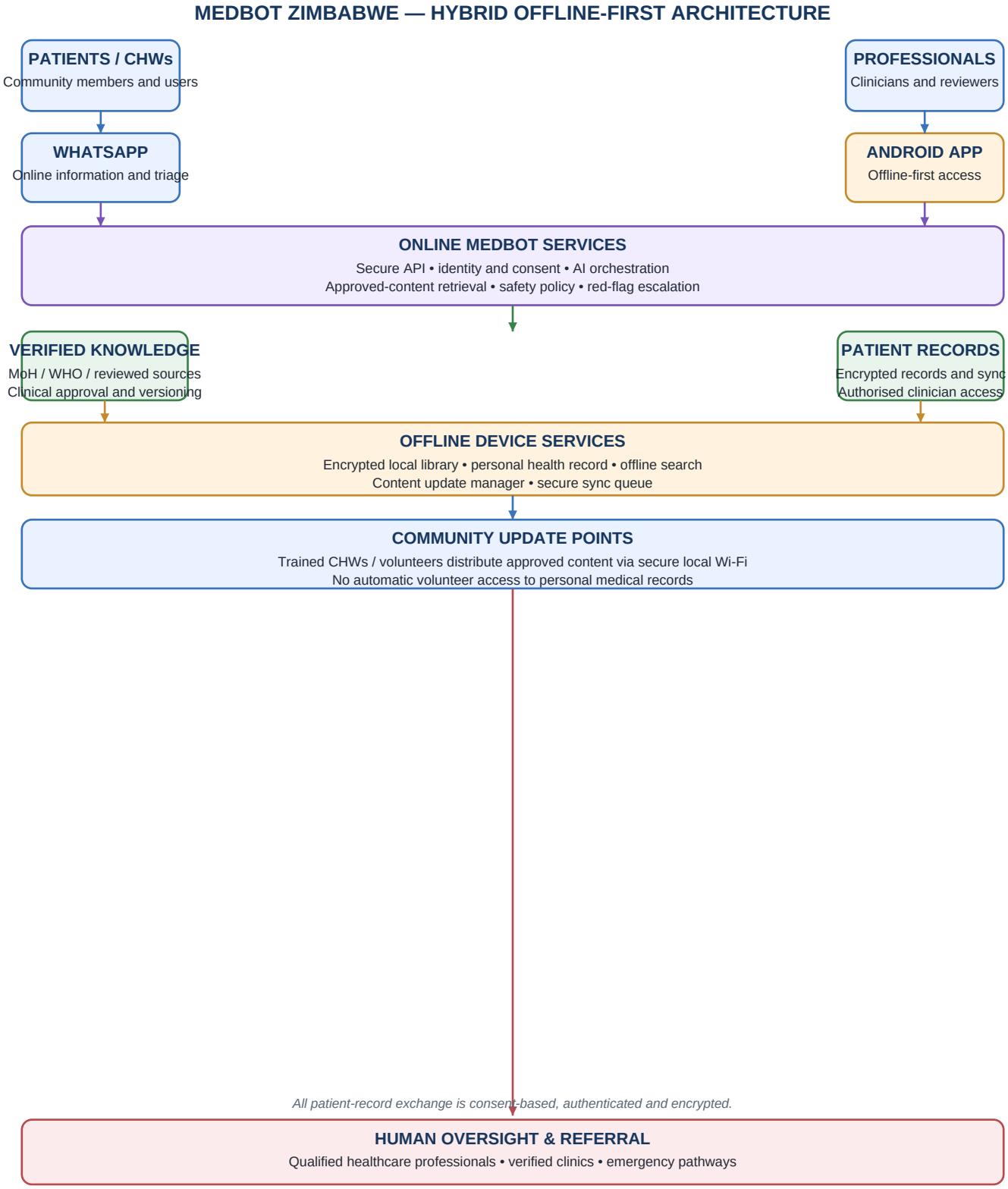

Figure 1. Conceptual system architecture. Arrows indicate high-level information flows; detailed interfaces and deployment boundaries are to be confirmed during technical design.

2. Architecture layers and responsibilities

Layer	Components	Responsibility
1. User access	Patients, CHWs, WhatsApp, Android app, clinician portal	Provides the interfaces through which users ask questions, access information and manage permitted records.
2. Online Medbot	Secure API, identity and consent, AI orchestration, retrieval, policy node	Processes connected requests, retrieves approved information, applies safety rules and escalates red flags.
3. Health knowledge	MoH / WHO sources, clinical review, versioned repository, signed packages	Controls provenance, review, approval, versioning and distribution of trusted health information.
4. Offline infrastructure	Encrypted local storage, health library, offline search, local records, sync queue	Keeps approved information and permitted records available on the device without internet.
5. Community network	Community Update Points, secure local Wi-Fi, trained CHWs / volunteers	Helps users receive approved content updates in areas with limited connectivity.
6. Patient records	Secure record service, synchronisation, audit logs, clinician dashboard	Synchronises permitted records and enables authorised professional access with appropriate consent.
7. Human oversight	Qualified professionals, clinics, verified emergency pathways	Provides clinical review, professional accountability and referral when needed.

3. Core technical workflows

A. Online healthcare consultation

Flow: Patient question → WhatsApp interface → secure API → identity / consent checks → AI orchestration → approved health-content retrieval → response generation → clinical safety policy → response or escalation.

The model should be grounded in approved sources. The policy layer must prevent diagnosis and unsupported treatment advice, and trigger urgent-care guidance for recognised red-flag symptoms.

B. Offline healthcare consultation

Flow: Android app → encrypted local content library → offline search / constrained explanation → answer with source and content version.

Offline functionality should initially prioritise search and retrieval from approved local content. Running a full language model on-device is a later option dependent on device capability and clinical validation.

C. Personal health record

Flow: Patient-controlled record → encrypted local storage → consent and access checks → authenticated synchronisation → secure record service → authorised clinician view.

Patients should be able to review and correct their information. Access should be limited to what is necessary, logged and governed by applicable law and clinical policy.

D. Community content update

Flow: Clinically approved content package → secure update device / local Wi-Fi → Community Update Point → user device → signature and version check → local library update.

Community volunteers may help distribute approved information but must not receive automatic access to personal medical records.

E. Store-and-forward synchronisation

Flow: Offline changes → local secure queue → trusted connection available → authentication and encryption → server validation → conflict handling → sync status and audit log.

The system should handle interrupted transfers, duplicate updates and conflicting edits without silently overwriting newer or clinically important information.

4. Recommended technology stack

The following options are recommendations for technical evaluation, not confirmed project selections.

Component	Potential technology	Reason / role
Android application	React Native	Cross-platform mobile development with Android support.
WhatsApp integration	WhatsApp Business Platform API	Connected messaging access for users with internet connectivity.
Backend services	Python FastAPI or Node.js	API layer connecting channels, AI services and data services.
AI and retrieval	LLM with retrieval-augmented generation (RAG)	Grounds responses in approved health information and supports controlled retrieval.
Offline database	Encrypted SQLite	Local storage for approved content and permitted user records.
Online database	PostgreSQL	Structured storage for authorised patient-record services and application data.
Content storage	Secure object storage	Stores approved documents and versioned content packages.
Authentication	OAuth 2.0 / OpenID Connect	Supports identity verification and access-control integration.
Synchronisation	Secure REST API and background sync	Transfers permitted changes when a trusted connection is available.
Clinician portal	React web application	Provides authorised professionals with a controlled interface to permitted records.

Technology choices should be confirmed after assessing device affordability, offline reliability, data-protection requirements, local developer capacity, hosting location and integration needs.

5. Security, privacy and clinical governance

Encryption

Encrypt sensitive records on devices and in transit. Protect encryption keys and avoid storing secrets in application code.

Authentication and authorisation

Verify users and professionals. Apply role-based and least-privilege access, with stronger verification for clinical users.

Patient consent and control

Provide clear consent and sharing controls, subject to applicable legal requirements. Support correction, revocation and appropriate access history.

Auditability

Log access, record changes, synchronisation events and administrative actions. Protect logs from unauthorised alteration.

Device and account protection

Plan for shared phones, lost devices, forgotten credentials, local session timeouts, secure backups and recovery.

AI safety

Do not diagnose, prescribe or independently change treatment. Use approved sources, display limitations and escalate urgent warning signs.

Clinical governance

Assign accountable Zimbabwean clinical reviewers, content owners, review dates, withdrawal processes and incident escalation responsibilities.

Young people and safeguarding

Apply age-appropriate confidentiality, safeguarding and child-protection procedures to sensitive topics and records.

Legal compliance

Before deployment, identify applicable Zimbabwean privacy, health-data, professional-practice and cross-border data-transfer requirements.

6. Phased implementation roadmap

Phase 1 — Foundation and governance

Confirm Ministry of Health, clinic, CHW and professional stakeholders. Agree clinical content ownership, applicable privacy requirements, consent, security, safeguarding, referral and incident-response procedures. Define the offline application requirements.

Phase 2 — Offline information prototype

Build an Android prototype with approved downloadable health-information packages, Shona and Ndebele support, offline search, content versioning and red-flag escalation. Test offline usability and content integrity.

Phase 3 — Patient-record prototype

Develop a limited My Health Record prototype with user-controlled access, encryption, audit logs and a small set of agreed record fields. Validate consent and access workflows before live clinical use.

Phase 4 — Synchronisation and community pilot

Test authenticated, encrypted store-and-forward synchronisation, conflict handling and Community Update Points. Conduct controlled tests with CHWs and participating clinics, including interrupted and low-connectivity scenarios.

Phase 5 — Evaluation and responsible expansion

Evaluate safety, language quality, reliability, privacy, usability and referral outcomes. After review, consider maternal health, youth health, nutrition, disaster-health content and the verified professional contributor network.

7. Key design decisions still required

• Which patient-record fields are essential for the first pilot, and who is legally responsible for the records?

• Which Zimbabwean health-data and privacy requirements apply, including to hosting and any cross-border transfers?

• Which clinical body or named roles approve content, and how often is each content category reviewed?

• How will a patient prove identity and consent when using a shared or replacement phone?

• How will a patient recover records after device loss without exposing them to volunteers or unauthorised parties?

• Which clinics and emergency referral contacts will be verified for the initial pilot districts?

• What minimum Android device specifications, storage capacity and update methods are realistic for target communities?

• What measurable safety, reliability, language and usability criteria must be met before expanding the pilot?

CORE ARCHITECTURAL PRINCIPLE

One healthcare knowledge system. Two access channels. Secure patient records. A connected network of healthcare professionals.

This architecture is a proposed design for discussion. Clinical validation, privacy and security assessment, stakeholder approval and controlled pilot testing are prerequisites to deployment.